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When does Endorsement Work? The Moderating Role of Endorser and Message Characteristics

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Abstract

Purpose: This study explores how both endorser and message characteristics moderate the effectiveness of influencer endorsements.

Design and methodology: We adopted a quasi-experimental design to track endorsement events on social media and corresponding sales data from e-commerce platforms. Difference-in-difference (DID) models and regression analysis were used to examine the effects of celebrity endorsements and the moderating effects of various endorser and message characteristics.

Findings: We find that only influencers with over 10,000 followers significantly drive sales. Additionally, frequent endorsements by the same influencer are less effective. In terms of message characteristics, imagery and overt endorsements are found to be more effective than their text-only and subtle counterparts.

Originality: This study contributes to the influencer marketing literature by revealing the role of endorser and message characteristics in influencing the effectiveness of online influencer endorsements, and by quantifying the impact of online influencer endorsements on actual sales.

Practical Implications: On average, influencers with less than ten thousand followers are unlikely to drive significant sales. Influencers need to be more strategic in their endorsement activities, as frequent endorsements can be detrimental. Overt and imagery endorsement messages are more effective than covert and text-only ones.

Keywords: online influencer endorsement, social media, e-commerce, field study

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1. Introduction

For decades, companies have adopted celebrity endorsement by hiring actors/actresses, singers, and athletes to promote their products through mass media (Erdogan 1999). While such practices are still prevalent today, many companies now complement them with social media strategies to reach younger consumers, who increasingly rely on social media for shopping decisions. As part of this shift, firms are partnering with online influencers to promote their products on platforms like Twitter, Instagram, TikTok, and Weibo (De Veirman et al. 2017, Hwang and Jeong 2016, Brison et al. 2016, Jin and Phua 2014, Yang and Ha 2021). The estimated global influencer marketing expense reached 30.81 billion USD with per internet user spending at 5.29 USD in 2023, and it is forecasted to reach 39.33 billion USD by 2025, according to a market research report by Statistia (Statista 2025).

Influencer marketing is "a strategy in which a firm selects and incentivizes online influencers to engage their followers on social media in an attempt to leverage these influencers' unique resources to promote the firm's offerings" (Leung et al. 2022b, pp. 1). This approach is similar to traditional celebrity endorsement strategies but is adapted to the social media context (Bergkvist and Zhou 2016). Online influencers promote a firm's offerings with the purpose of increasing consumers' purchase intentions and, ultimately, enhancing the firm's performance through endorsement activities conducted on social media platforms.

An online endorsement comprises two key components: the endorser, who serves as the source of the message, and the endorsement message itself. Scholars have examined various factors that are either from endorser or message perspective. Regarding the endorser , follower size remains a key factor for marketers when selecting influencers for marketing campaigns, as their effectiveness is largely driven by the influencers' fanbase (De Veirman et al. 2017, Jin and Phua 2014, Cascio Rizzo et al. 2023b). While influencer follower size has received substantial attention in the literature as a critical factor, findings remain mixed and inconclusive (Kay et al. 2020, Leung et al. 2022a, Park et al. 2021). For example, while Leung et al. (2022a) finds that follower size enhance endorsement effectiveness, Kay et al. (2020) finds the micro influencer – those with fewer followers are more effective in endorsing products. Beyond follower size, however, endorsement frequency — defined as the proportion of endorsements relative to an account's overall posting activity — has received limited attention in influencer marketing research. Meanwhile, endorsement message is equally important in determining its effectiveness. Past studies have examined the effects of the language used in an endorsement (Cascio Rizzo et al. 2023b, Cascio Rizzo et al. 2023a), the tone of the messages (Jin and Phua 2014), and the disclosure of endorsements (Hwang and Jeong 2016), while other factors such as the format of the message (i.e., text vs.

imagery) and the intent (overt vs. covert) have not yet been explored. In addition, despite the significant investments in influencer endorsements, their impact on actual product sales remains unclear. Existing research examining sales outcomes typically focuses on a single influencer type or a narrow product category (Beichert et al. 2024, Chung et al. 2013, Elberse and Verleun 2012), raising concerns about the generalizability of their findings. Expanding the scope of research to encompass diverse influencer types and product categories is crucial for generating more comprehensive understanding of influencer endorsement.

Recently, a growing number of studies have begun to address this gap by examining the sales effects of influencer endorsements using data from livestream platforms covering a wider range of products. For example, Gu et al. (2024) and Zhang et al. (2024) examine the actual sales impact of influencers in the context of livestream commerce, where influencers overtly promote products in a virtual showroom and engage directly with viewers to generate sales within the platform. However, our context differs from livestream studies in that we examine the sales impact in a cross-platform setting, where endorsements originate on social media but aim to drive purchases on external e-commerce platforms. The need for cross-platform navigation represents a form of cognitive burden that can potentially affect how consumers process information. As a coping strategy for the additional cognitive demands of platform switching, consumers might be more reliant on endorser and message variables that are less effortful to process. In addition, endorsement messages in this setting differ in their overtness (ranging from overt and explicit product promotion to subtle and implicit mentions) and their message types (text-only message or imagery with both text and visuals). These two features from endorsement message perspective are distinctive in cross-platform endorsement and reflect strategic choices by influencers. Yet, their effectiveness in driving sales remains underexplored. Thus, there is an imperative need to study when and how social media endorsements succeed in generating product sales across platforms and to provide practitioners with quantifiable guidelines for selecting appropriate influencer and message strategies.

In this study, we analyze the effect of online influencer endorsements on product sales using real-world data collected from both social media and e-commerce platforms. We examine six types of influencers, food and drink, beauty, makeup bloggers, celebrities, athletes, and sports. Unlike previous research that relies on experiments or surveys (De Veirman et al. 2017, Jin and Phua 2014) our study focuses on actual sales (behavioral response) rather than consumer attitudes and intentions (internal response). Specifically, our research has two objectives. First, we study the impact of online influencer endorsement on actual product sales using field data directly collected from social media and e-commerce sites for various types of influencers. Second, we explore when online influencers

would be more effective by exploring contingency factors including endorser (influencer) and message characteristics.

We use elaboration likelihood model (ELM) of persuasion as a lens and draw on the source credibility theory and the information processing theory to explain how endorser and message characteristics moderate the effectiveness of online influencer endorsements, respectively. Our analysis reveals several important findings. First, we did not find an average effect of online influencer endorsement on actual sales. This underscores the need to explore the specific circumstances under which online influencer endorsements are effective in driving sales. Our further analysis indicates that the characteristics of the endorser and the features of the endorsement message are crucial in the context of online influencer endorsement (Jin and Phua 2014). From the perspective of the endorser, the number of followers plays a significant role in determining the effectiveness of the endorsement. Our quantitative analysis reveals that, on average, every ten thousand followers can lead to an increase of about 6 units in monthly sales. Additionally, we have found that frequent endorsements have a negative impact on sales, suggesting the importance of endorsers being more strategic and cautious in their product endorsements. From the perspective of the endorsement message, we have discovered that imagery endorsements and overt endorsement messages generate more sales compared to text-only endorsements and covert endorsement messages, respectively.

Our study makes important contributions to both research and practice. First, our research provides field evidence on the effectiveness of online influencer endorsements by tracking endorsement events on social media and collecting actual sales data from the e-commerce platform. We thus complement existing lab-based studies that rely primarily on reported purchase intentions. Second, our study differs from prior research on livestream commerce by examining a cross-platform setting, where endorsements vary in both message type and overtness. Third, we provide a framework for understanding the factors that moderate the effectiveness of online influencer endorsements by incorporating the source credibility theory and the information processing theory. Fourth, we identify several endorser and message characteristics that moderate the effectiveness of the endorsement. We also reveal that the persuasion mechanism works differently in the social media context with the presence of information overload. Finally, we provide actionable endorsement strategies for practice.

The rest of the paper is organized as follows. We begin with an overview of online influencer endorsement in the literature. Next, we present the theoretical underpinnings of our study followed by our research framework and hypotheses. After that, we describe our field study design and present the empirical results. Finally, we discuss the theoretical and practical implications of this study, as well as its limitations.

2. Literature on Online Influencer Endorsement

2.1 Traditional Celebrity Endorsement

Previous research has indicated that the impact and success of celebrity endorsements are influenced by various factors. For example, source credibility has been shown to have a positive impact on the effectiveness of celebrity endorsements (Erdogan 1999). In particular, two components of source credibility – perceived level of expertise and trust in a celebrity – contribute to the effectiveness of the endorsement (Ohanian 1991, Roy Dholakia and Sternthal 1977). Another factor is the attractiveness of the endorser (Singer 1983). However, studies on the effect of attractiveness on brand evaluation and purchase intention have yielded mixed findings. While some studies have shown positive effects of celebrity attractiveness (Kahle and Homer 1985, Silvera and Austad 2004), others have found no effect (Fleck et al. 2012, Miller and Allen 2012), and even negative effects (Rossiter and Smidts 2012). In addition, one common limitation of previous studies is that the majority of the existing studies have mainly focused on the impact of celebrity attractiveness on consumer brand evaluation or purchase intention, rather than on the actual purchase behavior or product sales as a whole (De Veirman et al. 2017, Hsu et al. 2013, Jin and Phua 2014). Our study contributes to the traditional celebrity endorsement literature by presenting field evidence on the impact of celebrity endorsements on actual sales. We also explore how several endorser attributes and endorsement message characteristics moderate the effects of endorsements.

2.2 Online Endorsement Strategy

Online influencer endorsements differ from traditional celebrity endorsements in many ways. First, online influencer endorsements are generally easier and quicker to create than traditional endorsements, which involve extensive production and distribution processes. Online endorsements often take the form of social media posts containing text, photos, videos, or hyperlinks, which influencers can effortlessly create, edit, and share with their followers. Second, online influencer endorsements often appear as casual, personal experiences, making them seem less biased, more subtle, and ultimately more persuasive (De Veirman et al. 2017). In contrast, traditional celebrity endorsements are typically perceived as an overt persuasion process. Studies suggest that overt endorsements reduce message credibility due to their association with persuasive intent (Chan 2020, Hwang and Jeong 2016, Liljander et al. 2015, Waltenrath 2024). Third, while research on traditional endorsements has shown that excessive product endorsements by celebrities can harm credibility and consumer attitudes (Tripp et al. 1994),

online influencers face different dynamics. Lower content creation and distribution costs allow influencers to post endorsements more frequently. Furthermore, followers may view frequent endorsements as a norm in online contexts, potentially reducing the negative effects of endorsement frequency. However, whether these patterns hold true in online contexts remains underexplored.

Scholars have increasingly examined the effectiveness of online influencer endorsements, focusing on two key components: the endorser (the source of the message) and the endorsement message itself. As summarized in Table I, prior research has identified a range of endorser and message related factors that influence endorsement outcomes, including consumer attitudes, purchase intention, user engagement and sales. From the endorser perspective, follower size and source credibility dimensions have emerged as particularly prominent factors in shaping endorsement effectiveness. Follower size is often viewed as a key indicator of an influencer's popularity and reach, although findings have been mixed regarding whether larger follower size consistently generate greater impact (Beichert et al. 2024, Cascio Rizzo et al. 2023a, De Veirman et al. 2017, Gu et al. 2024, Jin and Phua 2014, Kay et al. 2020, Leung et al. 2022a, Park et al. 2021, Tian et al. 2024, Valsesia et al. 2020, Wies et al. 2023). For example, Kay et al. (2020) and Park et al. (2021) found that micro-influencers demonstrated higher effectiveness in affecting consumers' purchase intention in experimental settings. However, Leung et al. (2022a) finds that follower size enhance endorsement effectiveness using field data. Alongside follower size, source credibility dimensions, including perceived trustworthiness, attractiveness, expertise and reputation, have been consistently identified as critical determinants of consumer response (He and Jin 2022, Hsu et al. 2013, Liao et al. 2023, Lou and Yuan 2019, Masuda et al. 2022, Yuan and Lou 2020, Zhang et al. 2024). Other influential attributes include influencer-brand fit and emotional or relational cues like displayed emotions and perceived sincerity (Breves et al. 2019, Lee and Eastin 2020, Lin et al. 2021). From the message perspective, scholars have examined how specific features of the endorsement content affect outcomes. Sponsorship disclosure is one critical element, as it enhances transparency but can also provoke skepticism (De Veirman and Hudders 2020, Kim and Kim 2021, Leung et al. 2022a). Additionally, content attributes like hedonic or informational value, emotional tone, and sensory language for vividness have all been associated with differences in endorsement effectiveness (Cascio Rizzo et al. 2023a, Hughes et al. 2019, Jin and Phua 2014, Leung et al. 2022a, Lou and Yuan 2019).

However, most existing studies have emphasized engagement metrics, such as likes, shares, and comments, rather than actual economic outcomes, such as product sales, largely due to data availability. Leung et al. (2022a), for example, developed a conceptual framework of the effectiveness of online influencer endorsement, which encompasses influencer, receiver, and message characteristics. Using data collected from an influencer marketing

platform in China, they focused primarily on post engagement (i.e. repost). Cascio Rizzo et al. (2023b) examined how message characteristics (high vs. low arousal) interact with influencer characteristics (micro vs. macro) to impact message engagement. Wies et al. (2023) identified an inverted U-shaped relationship between follower size and engagement with endorsement messages. The lack of studies directly linking online endorsements to actual sales outcomes reflects a gap in both theoretical and empirical understanding of influencer endorsement strategies. Considering the substantial investments made by companies in online influencer endorsements, it is crucial to investigate the real impact of such endorsements on actual product sales.

-- Insert Table I Here --

2.3 Studies of the Economic Value of Endorsement on Sales

Studies on celebrity endorsement have traditionally measured effectiveness through brand evaluation (De Veirman et al. 2017, Spry et al. 2011, Silvera and Austad 2004) and purchase intention (Jin and Phua 2014, Hsu et al. 2013, He and Jin 2022) as indicators of effectiveness. There is a gap between intention and actual behavior (Morwitz et al. 2007, Infosino 1986). Empirically, it would be more meaningful to use product sales directly as a measure of celebrity endorsement effectiveness. A few studies have measured the effectiveness of celebrity endorsement on product sales. Elberse and Verleun (2012) analyzed 51 athlete endorsements, finding significant sales increases in 43 cases, with results varying based on the athletes' achievements. Similarly, Chung et al. (2013) found that Tiger Woods' endorsement significantly boosted Nike golf ball sales by attracting competitors' customers and increasing overall demand. However, these studies have largely been conducted in the traditional advertisement settings, limiting their applicability to online contexts.

The dearth of studies on actual sales led Bergkvist and Zhou (2016) to call for more research on the relationship between celebrity endorsements and sales across various types of celebrities. Recently, a few studies have begun to examine the actual impact of online endorsements on sales but mainly in the livestream context. Bharadwaj et al. (2022) found emotional display in livestream commerce have negative u-shaped effects on sales over time. Gu et al. (2024) studied how to choose a mix of large and small online celebrities (influencers) to maximize product sales. Huang and Morozov (2024) found that promoting games on livestream platforms has a short-lived positive impact on sales. Beichert et al. (2024) showed that using low-followership online influencers have a better return on investment. These studies have mostly focused on the influencers' followership, while our study examines

multiple messages and influencers' characteristics. Moreover, the livestream context differs substantially from our study setting. On social media platforms, endorsement messages may appear as text-only or as imagery posts that combine visuals and text, whereas endorsements in livestream commerce are delivered exclusively in video format. Furthermore, unlike livestream commerce, where influencers overtly promote products in a virtual showroom and directly encourage purchases, the overtness level of endorsement messages on Twitter-liked social media varies considerably. The unique features from this cross-platform context remains underexplored but is critical for understanding influencer marketing effectiveness outside of livestream. In the next section, we will discuss the theoretical foundations of our research.

3. Theoretical Foundations

An online endorsement comprises two key components: the endorser, who serves as the source of the message, and the endorsement message itself. Both types of information can play important roles in shaping consumer decisions (Borchers et al. 2022, Farivar et al. 2023, Silvera and Austad 2004). The influence of online influencer endorsements on consumer decision-making can be viewed as a persuasion process. In this way, the elaboration likelihood model (ELM) of persuasion (Petty et al. 1986) offers an useful lens. According to the ELM model, individuals can engage in thoughtful consideration of the arguments presented in persuasive messages (i.e., central processing), such as endorsement content, or they can rely on cues when making purchasing decisions (i.e., peripheral processing), such as endorser's credibility. We argue that the cross-platform setting, in which consumers must navigate from social media to e-commerce platforms to complete purchases, imposes additional friction that demands additional cognitive effort. This may prompt consumers to rely on cues that are easier to process, even when they initially engaged in effortful processing of the endorsement content (O'Keefe 2013).[1] Building on this, we employ the Source Credibility Model to investigate how endorser (influencer) characteristics moderate endorsement effectiveness, and the Information Processing Model to examine the moderating role of message characteristics.

[1] The ELM model also explains which type of information plays a more important role under different conditions. When consumers are motivated and able to process information, they are more likely to engage in the central route, carefully evaluating the endorsement message. In contrast, when motivation or ability is low, they tend to rely on peripheral cues, such as the credibility or attractiveness of the endorser, to form attitudes and make purchase decisions. These contingencies are beyond the scope of this research; we leave them for further exploration.

3.1 The Source Credibility Model

The Source Credibility Model highlights the significance of the perceived characteristics of the communication source (i.e., the endorser) on the recipient's perception and behavior. These characteristics include trustworthiness, expertise, and attractiveness (Erdogan 1999, Halder et al. 2021, Hovland et al. 1953, Ohanian 1991). When a celebrity endorser is perceived as trustworthy and knowledgeable, it tends to positively impact the recipient's attitude and behavior towards the endorsed brand or product. Meanwhile, *attractiveness* not only encompasses physical attractiveness but also other qualities that the celebrity possesses, like intellectual skills, personality traits, lifestyle, and athletic prowess (Erdogan 1999, Kahle and Homer 1985). These findings are also in line with the social influence theory, which suggests that individuals are more likely to conform to the opinions of those they like, feel familiar with, and perceive as similar (Kelman 2017). In this study, we examine the characteristics of the online influencers, using source credibility model as the underpinning mechanism.

3.2 Information Processing Theory

The Information Processing Theory is invoked to explain how the characteristics of the endorsement message affect the effectiveness of the endorsement. The Information Processing Theory suggests that people have limited cognitive capacity to process stimuli, a constraint often referred to as the cognitive bottleneck (Simon 1994). On social media, the abundance of information can easily exceed users' cognitive limits, resulting in information overload, a state in which the volume of information surpasses an individual's ability to effectively process it (Fu et al. 2020). Numerous studies have examined the impact of information overload on social media user behavior, highlighting its role in reducing engagement, increasing fatigue, and shaping information avoidance strategies (Feng et al. 2015, Fu et al. 2020, Zhang et al. 2023).

In the context of online influencer endorsement, information overload poses a challenge to endorsement effectiveness. When consumers experience information overload, their limited cognitive resources are further taxed, making them less likely to pay attention to endorsement messages and more inclined to focus on simpler, more salient cues. For example, they may attend to visual content rather than text-based messages, which require greater cognitive effort to process (Sanbonmatsu and Kardes 1988). This shift reduces the likelihood that detailed endorsement content will be carefully evaluated, thereby affecting consumers' attitudes and purchase decisions (Zhang et al. 2023).

The information processing perspective has been used in the literature to explain various consumer behaviors and perceptions (Tam and Ho 2006, Tybout et al. 1981, Zhang and Goh 2018). In this study, we adopt the Information

Processing Theory to explain how the characteristics of endorsement messages impact their diagnosticity in the context of online influencer endorsements. This is particularly relevant due to the prevalence of information overload on social media platforms (Feng et al. 2015, Fu et al. 2020).

4. Research Framework and Hypotheses Development

The effectiveness of celebrity and influencer endorsements are affected by various factors, primarily from two perspectives, endorser and endorsement message (De Veirman et al. 2017, Goldsmith et al. 2000, Hwang and Jeong 2016, Jin and Phua 2014, Lafferty and Goldsmith 1999). In this study, we use the Source Credibility Model and the Information Processing Theory to explain how the characteristics of the endorser and endorsement message can jointly influence behavioral response (actual purchasing behavior) in a cross-platform setting, where message type and message overtness represent the distinctive features. We propose that the effectiveness of the endorsement depends on various endorser (influencer) and message characteristics. This study differs from previous studies that focused on internal responses including attitudes or purchase intentions in celebrity endorsement literature. Figure 1. provides a summary of our research framework.

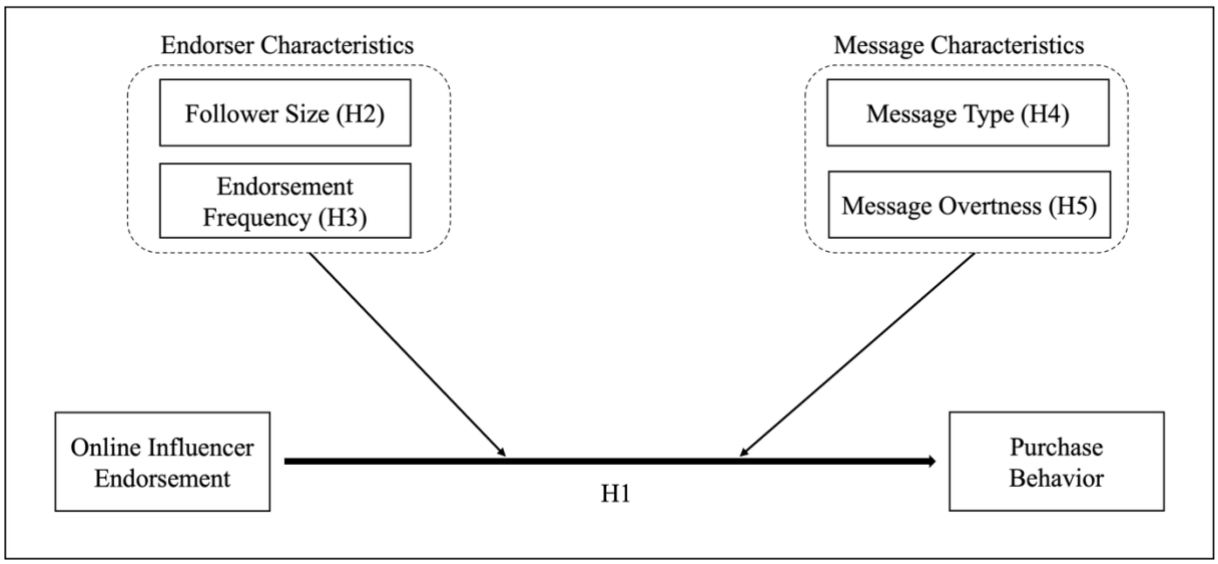


Figure 1. Research Framework

4.1 Online Influencer Endorsement

Consumers' behavioral responses to online influencer endorsements can be understood through the lens of persuasion theories in literature, including social influence theory, the elaboration likelihood model (ELM), and the stimulus-organism-response (SOR) model (Chen 2023, Kelman 2017, Mehrabian and Russell 1974, O'Keefe

2013). According to social influence theory, individuals are more prone to adopt the opinions of those they like, feel familiar with, and identify as similar (Kelman 2017). The ELM proposes that persuasion operates through two distinct routes: a central route, where motivated and able consumers carefully evaluate message content, and a peripheral route, where consumers with lower motivation or ability rely on cues such as the endorser's credibility or attractiveness (O'Keefe 2013). The SOR model further suggests that online influencer endorsements function as stimuli that evoke emotional and cognitive responses in consumers, which in turn shape their purchase behavior (Chen 2023, Mehrabian and Russell 1974).

Prior research has consistently demonstrated that both celebrity and influencer endorsements positively influence consumer attitudes, enhance purchase intentions, and even drive actual sales (Chung et al. 2013, De Veirman et al. 2017, Elberse and Verleun 2012, Jin and Phua 2014, Silvera and Austad 2004). Building on the above theoretical and empirical foundations, we hypothesize that online influencer endorsement on social media has a positive impact on product sales:

H1: Online influencer endorsement on social media significantly increases product sales.

4.2 Endorser Characteristics

Follower Size

Industry practitioners recognize that not all online influencers can effectively persuade consumers (Wies et al. 2023). This is in stark contrast to the case of celebrity endorsement, where the endorsers often possess broad recognition and established appeal. As such, marketing practitioners must carefully select influencers who can effectively engage their target audience (Gu et al. 2024). According to the Source Credibility Model, consumers are more likely to be persuaded by attractive influencers (Erdogan 1999, Kahle and Homer 1985, Spry et al. 2011). In the context of online influencer, follower size serves as a visible indicator of popularity and attractiveness (De Veirman et al. 2017). Moreover, source credibility has been found to influence attitude by affecting the own-thought activation process in cognitive response (Harmon and Coney 1982). A highly credible source inhibits own-thought activation, leading message receivers to rely on the information presented in the endorsement message, rather than making judgement basing on the information previously stored in one's memory (Greenwald 1968). Thus, when an influencer is perceived as more credible—reflected in a higher follower size—consumers are more likely to respond positively to the endorsement.

H2: The positive impact of online influencer endorsement on product sales is further enhanced when the online influencer has a larger follower size.

Endorsement Frequency

The perceived credibility of the online influencer can also be influenced by the endorsement frequency of the online influencer. Endorsement frequency is defined as the proportion of an influencer's social media posts that are product endorsements. Endorsement frequency is closely linked to the trustworthiness dimension within the Source Credibility Model. Prior research indicates that when celebrities endorse too many products, it undermines their perceived credibility (Tripp et al. 1994). This is because the frequent endorsements create the impression that the online influencer is extrinsically motivated by financial gain (Cheng et al. 2020). Therefore, we argue that product endorsement by an online influencer with a high frequency of endorsements leads to a lower level of perceived credibility, which weakens the impact of the endorsement on sales:

H3: The positive impact of online influencer endorsement on product sales is weakened when the online influencer's endorsement frequency is high.

4.3 Message Characteristics

Message Type

Endorsement messages on social media platforms can appear either as text-only or as imagery (i.e., text-and-image posts). Information processing theory suggests that individuals have a limited capacity to process incoming stimuli (Simon 1994). When the amount of information received exceeds the individual's processing capacity, the individual is likely to experience information overload (Lee and Lee 2004). In response, individuals may "compress, aggregate, categorize, and structure information" for easy processing (Eppler and Mengis 2004, pp. 335). Another coping strategy is to turn to less complex information like visuals (Chan 2001, Lewis et al. 2013), because images are more salient cues that can quickly capture the attention of the audience (Lin and Boh 2021). This is also consistent with attention economy theory, which suggests that individuals' attention is a limited resource (Davenport and Beck 2001, Falkinger 2008). The issue of limited information capacity is particularly salient in the context of social media. With the ever-increasing amount of information on social media platforms, users are less able to process and pay attention to each piece of information. Messages, including endorsement messages, appearing on social media platforms compete for users' attention (Falkinger 2008). Under conditions of information overload, consumers are more likely to process less complex information presented in a more

salient manner, such as visual messages, rather than textual messages which require more cognitive effort to process (Sanbonmatsu and Kardes 1988).

The presence of image in the endorsement message can be a salient cue that enhances the effectiveness of the endorsement for two reasons. First, the presence of an image might be more likely to capture viewers' attention. Early results from Pieters and Wedel (2004) using eye-tracking data on print advertisements have shown that image component can capture readers' baseline attention. Secondly, viewers may appreciate the extra effort put into creating imagery posts compared to text-only posts. This effort may serve as an additional cue for determining the credibility of the source (Kruger et al. 2004). Additionally, a study of image content and social media engagement found a significant positive effect of image presence on user engagement. The results were robust across different product categories and platforms (Li and Xie 2020). Hence, visuals in endorsement messages enhance the persuasive power of communication, particularly when recipients are less motivated to process detailed information. Once attention is secured, visuals complement the textual component by facilitating information processing rather than imposing additional cognitive demands. Miniard et al. (1991) suggest that imagery messages increase product evaluations when recipients are less motivated to process the information presented. This is due to the higher level of message diagnosticity perceived by recipients exposed to visuals compared to being exposed to text alone (Jiang and Benbasat 2007). Therefore, we argue that imagery endorsement messages incorporating both text and visuals are more effective in capturing consumer attention and enhancing message diagnosticity, thereby strengthening endorsement effectiveness:

H4: The positive impact of online influencer endorsement on product sales is strengthened when the online influencer endorsement is imagery.

Message Overtness

Unlike livestream commerce, where influencers overtly promote products in a virtual showroom and directly encourage purchases, the overtness level of endorsement messages on Twitter-liked social media varies considerably. We argue that the effectiveness of online endorsements is shaped by the degree of message overtness. Overtness refers to the extent to which consumers perceive the endorsement message as a deliberate and persuasive effort. Overt endorsement messages are clearly promotional and sponsored, while covert endorsement messages have a more subtle promotional intention (Chan 2020). From the source credibility perspective, covert endorsements are more likely to be perceived as non-sponsored casual sharing of a good product. The covert endorsement message thus is perceived to be more trustworthy in terms of the functionality and quality of the

product. This is consistent with the persuasion knowledge model, which argues that ambiguous cues (covert endorsement) are less suspicious of firms' manipulative intentions compared to salient cues (overt endorsement) (Kirmani and Zhu 2007). Therefore, covert endorsements are expected to be more effective in influencing consumer behavior than overt endorsements. Literature also found evidence that overt endorsement messages have a negative impact on purchase intention. For example, Liljander et al. (2015) found that overt marketing in blog posts had a detrimental effect on consumers' purchase intentions. In a laboratory setting, Hwang and Jeong (2016) found that simple disclosure of sponsorship, where a simple sponsorship message was provided in the blog post, like "Sponsored post", led to negative source credibility perceptions and message attitudes. Similarly, Chan (2020) found that covert promotions delivered through third parties were perceived as more believable, resulting in more positive evaluations of the promoted brand.

Nevertheless, the context of online influencer endorsements on social media presents a different dynamic. Unlike blogs or traditional media, social media platforms are characterized by severe information overload, where users are overwhelmed by a continuous stream of competing content. As discussed, because of individuals' limited capacity in information processing, social media posts are competing for users' attention. According to Information Processing Theory, individuals prefer to rely on simple and direct cues when the incoming stimuli exceed the information processing capacity as a coping mechanism (Simon 1994). In this environment, covert endorsements, though potentially more trustworthy, may fail to capture sufficient attention or clearly convey the promotional message. In contrast, overt endorsements convey a direct message that is much easier to process and understand, increasing awareness of the endorsed product and enhancing message diagnosticity. Thus, we argue that overt endorsements are more likely to attract user attention in the high-competition context, thereby strengthening endorsement effectiveness:

H5: The positive effect of online influencer endorsement on product sales is strengthened when the online influencer endorsement is overt.

5. Methodology

5.1 Data collection

Our data was collected during the period from September 28th, 2016 to October 31st, 2016. We collected both online influencer activities and the corresponding sales data. The online influencer endorsement data was collected

from Weibo, a Chinese social media site similar to Twitter and Facebook. Originally launched as China's Twitter, Weibo gradually incorporated features of Facebook by allowing users to post longer messages embedded with multimedia, such as images, hyperlinks, and videos. With its 300 million users and 130 million daily active users during the data collection period, Weibo was the dominant social media platform in China, making it an ideal choice for our research. Other platforms like TikTok (Douyin) and Xiaohongshu (Little Red Book or RED) were not as prominent at that time. The sales data was collected from Taobao and Tmall, both subsidiaries of Alibaba, the leading e-commerce company in China.

To ensure the generalizability of our findings, we chose to focus on a diverse range of online influencers across six categories that are more likely to endorse a product, including food and drink, beauty, makeup bloggers, celebrities, athletes, and sports. We extracted the top 200 influencers based on the number of followers from each category. Ten research assistants were recruited to assist with data collection. Each research assistant was assigned 240 accounts to track daily posts for one month, making sure each influencer was tracked by two research assistants. The research assistants were required to follow the assigned influencers on Weibo and read their posts each day. Any post that mentioned a product, whether through text or visuals, and regardless of the explicitness of the promotion, was recorded as an endorsement activity and submitted to our dedicated data collection system at the end of each day. Figure 2 provides an example of an endorsement activity that includes both text and image in the endorsement message.



Figure 2. An Influencer's Endorsement Activity

Once an endorsement was found, the research assistant would proceed to find the endorsed product. If there was a product link to Taobao or Tmall in the endorsement message, the research assistant would record the link in their submission in our data collection system. If there was no product link in the endorsement, the research assistants were required to search for the product name on Taobao and record the first product in the search results as the product link of an endorsement. Search results in Taobao include products sold on both platforms, Tmall and Taobao. We also asked the research assistant to identify a similar product from the recommendation list displayed on the product page as the control group in the DID analysis. The selection of similar products was based on product category, brand, store, and price. A similar product must meet three criteria: (1) it is in the same product category and sold by the same store; (2) it has the same brand; and (3) has a price within 20% of the endorsed product. The similar product represents a control product that is similar to the endorsed product in various ways (e.g., brand, functionality, etc.), except that it is a product not endorsed by the influencer on the social media platform.

Each day during the data collection period, research assistants were required to read all the posts posted on that day and report endorsement activities and product links before midnight that day. Data collection scripts were then automatically launched at midnight to extract data from product pages. Sales data, inventory information, ratings, buyer comments, and other product-related information were collected for all the endorsed products as well as the similar products. For each of the online influencer endorsement identified by our research assistants, we collected data from the product page for 30 days to monitor the change in sales.

We also used Weibo's API service to retrieve online influencers' Weibo account information including number of followers. Not all online influencer accounts have endorsement activities during the data collection period. Finally, 1311 endorsement activities were identified by our research assistants during the data collection period. 527 endorsement activities were removed due to incomplete data. Some of the endorsed products were removed from Taobao or Tmall before we completed the 30-day sales monitoring. In total, we collected 784 endorsement activities from 165 celebrities.

5.2 Measures

Dependent variable. We used 30-day sales data published on the product page to estimate the impact of the endorsements. Our dependent variable, monthly sales, is defined as the number of products sold in the past 30

days.[2] For each product endorsement *i* identified by our research assistant, we recorded the sales data we collected on Day 1 (immediately after the endorsement) as the monthly sales before the endorsement. We then recorded the sales data we collected on Day 30 as the monthly sales after the endorsement. For each product endorsement, we also used the similar product for comparison (aka, control group).

Endorser characteristics. We examine two endorser characteristics: follower size and endorsement frequency. Follower size of online influencers serves as a visible indicator of popularity and attractiveness (De Veirman et al. 2017). We measure *follower size* using the number of followers each influencer had at the time of the endorsement activity (Leung et al. 2022a). Endorsement frequency affects users' perceptions toward the endorsement (Tripp et al. 1994). We measure *endorsement frequency* as the ratio of the number of product endorsement posts to the total number of Weibo posts by the influencer in a 30-day window, as the total number of posts within a time period varies across online influencers. A higher endorsement frequency indicates that a greater share of the influencer's content during that period was dedicated to product promotion.

Message characteristics. We examine two message characteristics: message type and message overtness. Text-only and imagery messages differ in the demands they place on users' information processing capacity (Lin and Boh 2021, Miniard et al. 1991), while the overtness of the message influences users' perceptions of source credibility and message diagnosticity (Hwang and Jeong 2016, Liljander et al. 2015, Simon 1994). We use two dummy variables to measure message characteristics of the endorsements. *Is Text* is used to indicate whether the endorsement message contained only text or also contained pictures (1 if the message only contains text, and 0 if the message contains picture). To measure if an endorsement message is overt or covert, we hired two additional research assistants – who were not involved in the data collection task and were unaware of the purpose of our study. We developed a coding scheme (see Table II) based on two established frameworks: the typology of masked marketing practices (Petty and Andrews 2008) and the four-factor model of sponsorship transparency (Wojdynski et al. 2018). The typology outlines six types of disguised communication, which helped operationalize overt versus covert messages by aligning their characteristics, e.g., lack of apparent sponsorship for subtle categories versus clear promotional intent for overt ones, with our scale. The four-factor model further differentiates advertising messages into sponsorship transparent (overt) or non-transparent (covert) across four attributes: brand presence, sponsor clarity, disclosure, and lack of deception. Building on the two frameworks, our coding scheme ranges

[2] Total sales or daily sales data are not available on the product page.

from 1 (extremely subtle) to 5 (extremely overt), with subtle messages defined by the presence of a product without seeming like sponsored promotion, and overt messages identified by clear sponsorship alongside product presence. After a brief training session, the two research assistants independently coded the endorsement messages. Agreement between the two coders was 95.89%. We collapsed rating scores into a dummy variable (*Overt*) by assigning 1 to endorsement messages coded as "extremely overt" and assigning 0 to the rest.

-- Insert Table II Here --

Control variables. A list of control variables is included in our model. *Site* is a dummy variable indicating where the product link is from (1 for Tmall and 0 for Taobao). *Price* is the price of the product in Chinese yuan. We also controlled for number of images (*Num Images*) and words (*Num Words*) in the product description. Finally, we controlled for the product ratings in terms of their description (*Description Rating*, indicating whether the product is same as the description), service (*Service Rating*), and logistics (*Logistic Rating*).

5.3 Model

We use the following model to test H1:

$$\#\ Sales_{it} = \alpha + \beta \cdot Endorse_i + \gamma \cdot Post_t + \delta' \cdot Endorse_{it} \cdot Post_t + \theta' \cdot Z_{it} + \varepsilon_{it} \quad (1)$$

In the above model, $\#\ Sales_{it}$ is the monthly sales of product *i* on Day *t* (*t = 1* or *30*). $Endorse_i$ equals 1 if *i* is an endorsed product and 0 if *i* is a similar product that was not endorsed. $Post_t$ equals 1 if *t = 30*, and 0 if *t = 1*. The coefficient δ' captures the effect of the online influencer's endorsement. Z_{it} is a vector of control variables and θ' is a vector of the associated coefficients.

To test the moderation effect of endorser and endorsement message characteristics, we include endorser or message-related variables in the interaction term in the following form:

$$\#\ Sales_{it} = \alpha + \beta \cdot Endorse_i + \gamma \cdot Post_t + \delta' \cdot Var_i \cdot Endorse_{it} \cdot Post_t \cdot + \theta' \cdot Z_{it} + \varepsilon_{it} \quad (2)$$

where Var_i is a vector of endorser or message-related variables. For endorser characteristics, we use follower size (*Num Followers,* in thousands) to test H2 and endorsement frequency (*EF*) to test H3. For message characteristics, we use dummy variables *Is Text* and *Overt* to test H4 and H5, respectively.

6. Empirical Results

6.1 Main Results

Table III provides summary statistics for the main dependent variables and the control variables. Table IV shows the empirical results. Model 1 examined the average effect of celebrity endorsement. We did not find significant average effect of celebrity endorsement, as the interaction term $\text{Endorse} \times \text{Post}$ is not significant ($\text{Endorse} \times \text{Post} = -1089.37, \text{n.s.}$). Therefore, we did not find support for Hypothesis 1. This result is consistent with the industry observation that not all online celebrities have the ability to effectively increase sales. It suggests that the impact of celebrity endorsement on sales is not universal and depends on various factors, such as the characteristics of the endorser and the message being conveyed.

From Model 2 to Model 5, we examined different sets of endorser and message characteristics to test H2 to H5. In Model 2, we found a significant effect of the endorsement, when the number of followers was included in the model ($\text{Endorse} \times \text{Post} \times \text{Num Followers} = 0.61, \text{p} < 0.01$). This finding supports H2, indicating that the number of followers has a significant influence on the effectiveness of celebrity endorsement. Additionally, we were able to quantify the economic value of the followers. Our estimation revealed that an increase of ten thousand followers could lead to an increase of 6 units in sales. Considering that the average number of followers in our dataset is approximately 1,500 thousand, this number holds substantial meaning and significance. In Model 3, we found that as the online influencer increases the promotion frequency, the effect of endorsement is weakened ($\text{Endorse} \times \text{Post} \times \text{Endorsement Frequency} = -385.99, \text{p} < 0.001$). H3 is supported. One percent increase in endorsement frequency results in a decrease of more than 300 units of monthly sales.

In Model 4 and 5, we found significant findings regarding the effectiveness of imagery messages and overt messages in increasing sales. Firstly, we observed that imagery messages had a much stronger impact. The interaction effect in Model 4 was negative and significant ($\text{Endorse} \times \text{Post} \times \text{Is Text} = -5680.37, \text{p} < 0.001$). This suggests that incorporating pictures in the messages significantly enhances their effectiveness in increasing sales. In Model 5, we found that overt endorsement messages strengthened the endorsement effect. The interaction effect was positive and significant ($\text{Endorse} \times \text{Post} \times \text{Overt} = 2078.83, \text{p} < 0.05$). Specifically, extremely overt endorsement message corresponds to a higher monthly sales number by about 2000 units on average. H5 is supported.

-- Insert Table III Here --

-- Insert Table IV Here --

6.2 Robustness Check with Matched Sample

To test the robustness of our results, we further employed Coarsened Exact Matching (Iacus et al. 2012) to match the treated group (endorsed products) and control group (unendorsed products) on a set of controls variables (Num Images, Num Words, Desc Rating, Service Rating, and Logistic Rating). CEM coarsen each variable into broader categories or bins. After coarsening, CEM performs exact matching on the coarsened variables. Once the matching is completed, the original values of the matched data are used for further analysis. We followed the default Sturges' rule when performing the matching (Ozer et al. 2023). We have 107 matched strata out of 237 strata, resulting in a matched sample of 2,625 observations. Table V shows the multivariate L1 distance before and after the matching. As we can see, the CEM have significantly reduced the imbalance between the treatment and control group, especially for the rating variables (38% reduction of the L1 distance), indicating the effectiveness of the matching. We then used this matched sample to re-estimate the models. The estimated results are presented in Table VI and are consistent with the main results.

-- Insert Table V Here –

-- Insert Table VI Here –

7. Robustness Test using Comments as a Proxy of Sales

One of the challenges to our main analysis is that we relied on monthly sales data on day 1 and day 30, because Taobao/Tmall only reports monthly sales data rather than total sales data or daily sales. As a robustness test to the main analysis, we turn to the cumulative number of comments on the product pages in Taobao/Tmall as a proxy for the cumulative sales number. Taobao/Tmall only allows buyers who complete the transactions to post comments for the product they have purchased. If we assume that the fraction of the buyers who post comments after purchasing the product remains relatively stable, then the number of comments is proportional to the sales number.[3]

In the robustness test, our focus is on the daily growth of the cumulative comments. We assume that the daily growth of the comments is influenced by a similar set of control variables. Additionally, we hypothesize that there

[3] Some media have reported that about 1~2 percent of buyers would post review comments. Our analysis would not be affected by the actual percentage as long as the percentage is relatively stable over time.

is a difference between the endorsed group and the control group in terms of the daily growth of comments. Specifically, the model is:

$$ln(Growth_{\{it\}}) = \beta_0 + \beta_1 endorse_i + \boldsymbol{\beta'_d}\, D_{it} + \boldsymbol{\beta'_c Control_{it}} + \mu_{i,t} \qquad (3)$$

where $Growth_{\{it\}} = Comments_t / Comments_{t-1}$, is the comment growth rate of sample *i* at day *t*. $endorse_i$=1 indicates the sample i is an endorsed product. D_{it} is a 28×1 row[4] vector of dummy coding with only the 1 in the *t* element and $\boldsymbol{\beta'_d}$ is a vector of the corresponding coefficients. $\boldsymbol{Control_{it}}$ is a vector of controls similar to the main analysis and $\boldsymbol{\beta'_c}$ is a vector of the corresponding coefficients. $\mu_{i,t}$ is the error that is not captured by the model.

Table VII shows the results for the robustness test. Model 1 is the baseline model that tests the impact of endorsement on the comment growth rate. Model 2 incorporates control variables in addition to endorsement. In Model 3 we interact endorsement and day index dummies. From Model 1, we observe that the coefficients on the day index dummies are negative, indicating a declining daily sales volume over time. It is difficult for products to maintain stable sales over time due to competition or other factors such as seasonal effects. The effect of the endorsement is significantly positive ($\text{Endorse} = 0.0096, \text{p} < 0.001$), indicating the effectiveness of online influencer's endorsement. Model 2 shows the similar findings when the control variables are included.

In Model 3, the interaction between endorsement and each day index dummy was examined. We only observed one significantly positive coefficient on day 4 ($\text{Day_Index} = 4 \times \text{Endorse} = 0.015, \text{p} < 0.1$). This indicates that there was a noticeable increase in the number of comments on day 4 after the online influencer endorsement. This finding is reasonable, as it is often expected that the impact of sales, reflected by the number of comments, may be delayed.

-- Insert Table VII here --

8. Discussions and Implications

While prior research has focused on offline and online celebrity endorsements using purchase intention as a proxy, our study aims to examine the impact of online influencer endorsement on actual purchase behavior, with a focus

[4] The first day is the base period so it does not have the growth rate data. For dummy coding, Day 2 is our based period, leaving us with only 28 dummies.

on endorsers and messages characteristics. We find no direct association between endorsements and sales, highlighting the importance of these characteristics in the online influencer endorsement process. We draw on the source credibility model and information processing theory to theorize and empirically test how these characteristics moderate endorsement effectiveness.

First, we've identified the follower size as a critical determinant of the effectiveness of an endorsement. This confirms that attractive endorsers are more effective in persuading potential consumers to purchase the product, because consumers view the follower size as an indicator of the source's credibility (Jin and Phua 2014). We further quantified the effect of follower size and found that for every ten thousand additional followers, product sales could increase by approximately six units. Academic research and industry reports have highlighted distinctions among influencer tiers (Cascio Rizzo et al. 2023b). Online influencers with fewer than ten thousand followers have been described as the nano-influencers (Lee and Junqué de Fortuny 2022). Our research provides some empirical support for this cutoff that micro-influencers are those likely to lead to material sales. We also address mixed findings in the impacts of micro- and macro- influencers on endorsement effectiveness. While experiments suggest micro-influencers excel at increasing purchase intention (Park et al. 2021, Kay et al. 2020), our results indicate macro-influencers are more effective at generating actual sales in a social media context. These insights have practical implications for businesses, offering guidance on optimizing influencer marketing strategies and assessing the cost-effectiveness of different influencer tiers for product promotion.

Second, we find that frequent endorsement of products by an online influencer reduces the effectiveness of the endorsements. This aligns with the Source Credibility Model, as excessive endorsements diminish the endorser's credibility. It is also consistent with recent findings that repeated endorsements by online influencers have little impact on product sales (Geng et al. 2024). Thus, endorsers face a trade-off: endorsing more products may boost short-term income but can harm their credibility, potentially jeopardizing future opportunities. This finding also highlights the importance for companies to consider an influencer's recent endorsement frequency, alongside their follower size, when selecting endorsers.

Third, our findings also highlight the significant role of information overload in the mechanism of the online influencer endorsements. Given the prevalence of information overload on social media and the dominance of heuristic decision-making, we find that imagery and overt endorsements are more effective than text-only and covert endorsements, respectively. According to Information Processing Theory, individuals tend to rely on simple and easy-to-process information when making decisions as a coping mechanism. Contrary to the findings of Hwang and Jeong (2016) and Chan (2020), we discovered that overt messages are indeed more effective in

persuading viewers to purchase products. We attribute this finding to the unique environment of social media, where information overload is more pronounced. In the context of online influencer endorsements on social media platforms, endorsement messages compete for consumer attention. Overt endorsements deliver a direct message that is easier to process and understand, thereby enhancing product awareness. These findings also speak to the growing literature and practice of stealth marketing. One of the reasons marketers engage in stealth marketing is that consumers react negatively to overt marketing (Liljander et al. 2015). However, our results suggest that this concern may be less relevant in the social media environment, where individuals simply process the endorsement messages as cues. In this process, the awareness effect of stealth marketing, which informs consumers about the existence of the product, outweighs the persuasive effect, which conveys about the trustworthiness of the endorsement. Consequently, direct and easy-to-process marketing messages are actually more effective in persuading consumers.

Our study presents significant theoretical implications. Firstly, we propose a framework to investigate the moderating factors that influence the effectiveness of online influencer endorsements on actual sales, considering both endorser and message characteristics. Our findings reveal that the persuasion mechanism differs in the social media context due to information overload. Unlike traditional mass media, where consumers encounter limited content, social media platforms overwhelm users with vast information. This phenomenon alters the way consumers process and respond to influencer endorsements, necessitating a re-evaluation of existing persuasion models. Our study contributes to this re-evaluation by demonstrating the nuanced effects of information overload on consumer decision-making processes. Secondly, we quantify the impact of online influencer endorsement on actual sales (behavioral response), ranging from micro- to macro-influencers. This addresses a critical gap in literature, where previous studies have primarily examined the effects of celebrity/online influencer endorsement on consumer attitudes and purchase intentions (internal responses) in lab or survey settings.

Our work differs from and complements existing studies on livestream commerce (Beichert et al. 2024, Gu et al. 2024, Zhang et al. 2024), where influencers promote products on platforms that enable seamless in-stream purchases. By contrast, our cross-platform setting requires viewers to navigate from a social media site to an e-commerce site, imposing extra cognitive effort on decision making. Under such conditions, consumers are more likely to rely on salient cues when making decisions, consistent with the ELM model. Our empirical results strongly support this previously unexplored argument. In addition, while prior livestream studies typically examine either endorser or message characteristics in isolation, our study considers both, incorporating message type and message overtness as distinctive features. Finally, our design allows for a direct comparison of sales

outcomes between endorsed and non-endorsed products, providing stronger evidence of endorsement effectiveness.

Our study also offers several practical implications for marketers and firms that invest heavily in influencer campaigns. By analyzing data from the field, we provide robust evidence on how influencer endorsements translate into measurable sales outcomes, thereby increasing the managerial relevance of our findings. First, our results suggest that online celebrity endorsement alone is not a guarantee of sales impact. In particular, influencers with fewer than ten thousand followers are unlikely to generate substantial sales. This indicates that marketers should carefully evaluate the cost-benefit trade-off when collaborating with micro-influencers. Second, endorsement frequency emerges as a critical strategic consideration. Our findings show that when influencers dedicate too large a share of their posts to endorsements, whether for the same or different product, their credibility diminishes, reducing the effectiveness of the endorsements. Marketers should therefore aim for a balanced approach that safeguards authenticity and prevents audiences from perceiving influencers as overly commercialized. Third, message characteristics play an equally important role. We find that overt, imagery endorsement messages outperform covert, text-only ones in driving sales. This implies that clear, salient, and visually engaging promotions are more effective in capturing attention and motivating purchase decisions. For practitioners, this means that collaboration with influencers should not only focus on who delivers the message but also on how the message is presented. By integrating both endorser- and message-level strategies, marketers can better allocate resources, improve return on investment in influencer marketing, and achieve stronger sales outcomes.

Beyond marketers, our findings also carry implications for platforms, influencers, and consumers. Platforms can foster healthier engagement by enhancing transparency through clear labeling of endorsements, providing analytics to help manage posting balance, and discouraging excessive commercialization through algorithm design. Influencers need to balance promotional and non-promotional posts and present endorsements authentically to preserve long-term audience trust. Consumers benefit when endorsements are transparent. Clear, overt messages help them make more informed decisions, while a balanced volume of promotions reduces persuasion fatigue and preserves the trustworthiness of influencer relationships. Together, these practices contribute to a more sustainable influencer marketing ecosystem that balances commercial effectiveness with authenticity and consumer trust.

9. Limitations and Future Research

Our study has several limitations. First, our empirical analysis is constrained by data availability. The Taobao/Tmall platform only discloses sales data over a rolling 30-day window, which prevents us from observing daily sales fluctuations and capturing short-term dynamics. To partially address this limitation, we conducted a robustness check using the daily number of product comments as a proxy for sales, which revealed that the endorsement effects are most prominent in the initial days following the endorsement. Moreover, the DID design in this study includes a control group of similar products that did not receive endorsements during the study period, which helps to isolate the endorsement effect from other confounding influences such as seasonality and marketing trends. Nonetheless, future research could benefit from access to more granular sales data or alternative platforms to further validate and extend our findings. Second, despite training research assistants extensively to identify endorsed products on Taobao/Tmall, some products may not have been accurately linked to the influencer's endorsement. If this occurred, it would have likely minimized the observed effects, as non-endorsed products would not have seen a sales boost. Third, our research focused on endorsements on a Twitter-like social media platform, where content is primarily text- and image-based. With the rise of short-video platforms like TikTok in recently years, we question whether our findings can be generalized to these platforms, where video content is more eye-catching yet more time- and cognitively- intensive to process. Existing research on video-based platforms largely centers on livestream commerce, which involves direct product promotion. Future research could extend our work by examining the effectiveness of online influencer endorsements in short-video formats which embed overt or covert promotional messages within entertainment or informational content, rather than relying on direct product promotion. Understanding how the level of message explicitness in these short videos influences sales could yield valuable insights into the evolving strategies of influencer marketing. Fourth, while we analyzed endorsement activities from six types of influencers, our dataset was not exhaustive due to the vast number of online influencers and the manual effort required for identification. While we believe the theoretical framework is applicable across influencer types, future research could investigate how characteristics of different influencer types and product categories moderate endorsement effectiveness. For example, research might explore how influencers' hedonic versus utilitarian appeal, or the hedonic and utilitarian attributes of products, influence consumer decision-making in the context of influencer marketing.

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Table I Review of Online Influencer Endorsement Literature

Authors	Context	Endorser attributes	Message attributes	Outcomes	Methods
Beichert et al. (2024)	Instagram	Follower size	Nil	Return on investment	Empirical
Bharadwaj et al. (2022)	Livestream	Nil	Face presence; emotional displays	Sales (within platform)	Empirical
Breves et al. (2019)	Instagram	Influencer-brand fit	Nil	Perceived credibility; brand evaluation; behavioral intention	Survey
Cascio Rizzo et al. (2023a)	Multiple social media platforms	Follower size	Sensory language	Engagement; purchase intention	Empirical and Experiment
Cascio Rizzo et al. (2023b)	Multiple social media platforms	Follower size	Language arousal	Engagement	Empirical and Experiment

De Veirman et al. (2017)	Instagram	Follower size	Nil	Influencer likeability	Experiment
De Veirman and Hudders (2020)	Instagram	Source credibility	Sponsorship disclosure	Brand attitude	Experiment
Gu et al. (2024)	Livestream (TikTok)	Follower size; Number of influencers;	Nil	Sales (within platform)	Empirical
Guo et al. (2024)	Livestream (Taobao)	Nil	Cognitive social call to actions; Affective social call to actions	Sales (within platform)	Empirical
He and Jin (2022)	Social media users	Influencer attractiveness; expertise; trustworthiness	Nil	Purchase intention	Survey
Hsu et al. (2013)	Social media users	Influencer trustworthiness; reputation	Nil	Purchase intention	Survey
Huang and Morozov (2024)	Livestream (Twitch)	Nil	Nil	Video game usage	Empirical

Hughes et al. (2019)	Motherhood	Influencer expertise	Hedonic value	engagement	Empirical
Jin and Phua (2014)	Social media users	Follower size	Post valence	Purchase intention	Experiment
Kay et al. (2020)	Social media users	Follower size	Sponsorship disclosure	Purchase intention	Experiment
Kim and Kim (2021)	Social media users	Nil	Sponsorship disclosure	Product attitude	Experiment
Lee and Eastin (2020)	Instagram	Perceived sincerity	Nil	Attitude toward influencer; brand attitude; Purchase intention	Experiment
Leung et al. (2022a)	Weibo	Influencer activity; influencer originality; follower size	Post positivity; sponsor salience; new product launch	engagement	Empirical
Liao et al. (2023)	Socia media users	Influencer expertise	Nil	Purchase intention	Survey

Lin et al. (2021)	Livestream	Influencer emotion	Nil	Engagement	Empirical
Lou and Yuan (2019)	Social media users	Trustworthiness; attractiveness; similarity to followers	Informative value	Perceived trust; brand awareness; purchase intentions	Survey
Masuda et al. (2022)	Social media users	Influencer attractiveness; expertise; trustworthiness	Nil	Purchase intention	Survey
Park et al. (2021)	Social media users	Follower size	Hedonic vis utilitarian consumption	Perceived influencer authenticity	Experiment
Tian et al. (2024)	Livestream (TikTok)	Follower size	Video features: informational, entertaining, socializing	Engagement	Empirical
Valsesia et al. (2020)	Twitter	Follower size; number of followees; perceived autonomy; perceived influence	Nil	Engagement	Experiment

Wies et al. (2023)	Instagram	Follower size	Content customization	Engagement	Empirical and Experiment
Yuan and Lou (2020)	Social media users	Perceived credibility; perceived fairness	Nil	Product interest	Survey
Zhang et al. (2024)	Livestream (Diantao)	Influencer likeability; influencer expertise	Nil	Sales (within platform)	Empirical
This study	Weibo and Taobao/Tmall	Follower size; Endorsement frequency	Message Format; Overt vs Convert	Sales (cross-platform)	Empirical

Table II Coding scheme for endorsement message

From 1 to 5: extremely subtle to extremely overt	
1: Extremely subtle;	
2: Moderately subtle;	
3: Moderate;	
4: Moderately overt;	
5: Extremely overt	
Subtle:	**Overt**:
1. Has a product	1. Has a product
2. Does not seem to be a sponsored promotion	2. Appear to be a sponsored promotion
3. Might induce viewers to buy the product	

Table III Summary statistics of the variables

	Endorsed				Control			
	Before		After		Before		After	
	Mean	S. D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
Num Sales	8157.89	18775.19	7263.20	18569.33	2167.52	9205.95	2347.20	10027.59
Price	445.47	998.08	428.06	875.44	419.27	890.07	419.10	899.41
Num Comment	17.92	52.33	20.41	56.18	9.27	43.95	9.10	43.65
Num Images	11.36	14.84	14.88	15.29	11.44	14.74	14.38	16.57
Num Words	149.18	456.60	164.49	484.89	135.28	400.16	145.39	402.12
Desc Rating	4.82	0.07	4.82	0.07	4.82	0.06	4.82	0.06
Service Rating	4.80	0.07	4.80	0.07	4.81	0.06	4.80	0.06
Logistic Rating	4.79	0.07	4.79	0.07	4.79	0.06	4.79	0.07
Num Followers	1592.40	2508.47						
Endorsement Frequency	3.78	4.84						
Is Text	0.88	0.32						
Overt	0.65	0.48						

Notes: Summary statistics of Number Followers (in thousands), Endorsement Frequency, Is Text and Overt of the endorsed products are only available for the endorsed group and reported under "before" columns.

Table IV Regression Results

	Model 1	Model 2	Model 3	Model 4	Model 5
Site	5701.43***	5621.46***	5621.46***	5729.22***	4756.83***
	(695.22)	(694.67)	(694.67)	(693.62)	(711.41)
Price	-1.56***	-1.53***	-1.53***	-1.55***	-1.45***
	(0.28)	(0.28)	(0.28)	(0.28)	(0.28)
Num Images	2.93	1.95	1.95	-0.62	-1.04
	(18.36)	(18.33)	(18.33)	(18.30)	(18.21)
Num Words	-0.10	-0.07	-0.07	-0.07	-0.03
	(0.63)	(0.63)	(0.63)	(0.63)	(0.63)
Desc Rating	21456.15***	21198.63***	21198.63***	21060.50***	19412.83***
	(5536.35)	(5528.83)	(5528.83)	(5516.25)	(5496.98)
Service Rating	-32929.36***	-31571.12***	-31571.12***	-30484.53***	-26695.62***
	(7875.06)	(7877.10)	(7877.10)	(7863.90)	(7854.12)
Logistic Rating	4187.41	3094.40	3094.40	2854.44	127.83
	(6693.87)	(6694.57)	(6694.57)	(6679.49)	(6664.30)
Endorse	6035.98***	4997.84***	4997.84***	5689.77***	6758.03***
	(722.44)	(533.65)	(533.65)	(560.87)	(589.35)
Post	172.73	-867.66	-867.66	-170.41	890.45
	(724.22)	(537.58)	(537.58)	(565.03)	(592.86)
Endorse × Post	-1089.37				
	(1021.05)				
Endorse × Post		0.61**	0.66***	0.66***	0.62**
× Num Followers					
		(0.19)	(0.19)	(0.19)	(0.19)
Endorse × Post			-385.99***	-280.51**	-313.91**
× EF					
			(98.39)	(99.67)	(100.66)
Endorse × Post				-5680.37***	-6010.63***

× Is Text					
				(1006.60)	(1016.15)
Endorse × Post					2078.83*
× Overt					
		`			(905.54)
Constant	33920.57	34439.06	30655.31	33498.73	35065.57
	(22929.24)	(22892.32)	(22860.14)	(22753.69)	(22748.39)
N	3136	3136	3136	3136	3136
Adjusted R^2	0.104	0.107	0.111	0.119	0.121

Notes. (1) *Standard Error* in parentheses. (2) *Num Followers* are in thousands. (3) Significance codes: * $p <$ 0.05, ** $p < 0.01$, *** $p < 0.001$

Table V Coarsened Exact Matching (CEM) Results

Matching Variable	Before Matching	After Matching
Num Images	0.063	0.067
Num Words	0.038	0.030
Desc Rating	0.223	0.000
Service Rating	0.223	0.000
Logistic Rating	0.082	0.000
Multivariate L1 Distance	0.430	0.267

Table VI Regression Results after Coarsen Exact Matching

	Model 1	Model 2	Model 3	Model 4	Model 5
Site	5788.25***	5682.80***	5845.24***	4725.78***	4645.18***
	(853.29)	(852.57)	(851.74)	(876.75)	(877.32)
Price	-1.53***	-1.50***	-1.52***	-1.40***	-1.42***
	(0.31)	(0.31)	(0.31)	(0.30)	(0.30)
Num Images	29.39	27.90	24.44	25.85	24.85
	(24.10)	(24.06)	(24.03)	(23.92)	(23.91)
Num Words	-2.07+	-2.02	-1.83	-1.53	-1.43
	(1.25)	(1.25)	(1.24)	(1.24)	(1.24)
Desc Rating	20021.44**	19584.02**	19168.66**	18534.50**	18565.94**
	(6294.93)	(6287.41)	(6273.72)	(6246.23)	(6243.08)
Service Rating	-36599.60***	-34817.15***	-33707.53***	-30198.89**	-30660.75***
	(9242.31)	(9249.65)	(9233.00)	(9217.28)	(9215.76)
Logistic Rating	8949.04	7522.08	7426.00	3753.55	3976.98
	(7863.34)	(7866.66)	(7848.28)	(7846.56)	(7843.45)
Treat	6506.38***	5258.64***	6046.40***	7192.38***	6670.19***
	(888.05)	(630.84)	(665.42)	(700.71)	(751.72)
Post	172.44	-1135.02+	-302.45	858.97	315.14
	(911.48)	(637.54)	(675.80)	(711.48)	(765.89)
Treat × Post	-1281.26				
	(1209.15)				
Treat × Post × Nun Followers		0.63**	0.67**	0.64**	0.60**
		(0.21)	(0.21)	(0.21)	(0.21)
Treat × Post × EF			-391.07***	-292.03**	-318.06**
			(107.27)	(108.59)	(109.39)
Treat × Post × Is Text				-5732.57***	-6037.46***
				(1143.56)	(1154.05)
Treat × Post × Overt					1943.93+
					(1016.65)
Constant	35282.77	36480.03	33059.83	36849.24	38218.68
	(25970.97)	(25923.28)	(25879.57)	(25771.99)	(25768.85)
N	2625	2625	2625	2625	2625
Adjusted R^2	0.101	0.104	0.108	0.116	0.117

Notes. (1) Standard Error in parentheses. (2) Num Followers are in thousands. (3) Significance codes: + p<0.1 * p < 0.05, ** p < 0.01, *** p < 0.001

Table VII Additional Analysis Results

	Model 1	Model 2	Model 3
Endorse	.0096***	.0073***	.0069
	(.0011)	(.0011)	(.0062)
day_index=3	-.011**	-.0057	-.0062
	(.0043)	(.0044)	(.0061)
day_index=4	.0065	.0047	-.0029
	(.0043)	(.0044)	(.0061)
day_index=5	-.01*	-.0023	-.0037
	(.0043)	(.0044)	(.0061)
day_index=6	-.0065	.000073	-.0021
	(.0043)	(.0044)	(.0061)
day_index=7	-.014***	-.0041	-.0029
	(.0043)	(.0044)	(.0061)
day_index=8	-.011*	-.000036	-.0021
	(.0042)	(.0044)	(.0062)
day_index=9	-.016***	-.0038	-.008
	(.0042)	(.0044)	(.0061)
day_index=10	-.018***	-.0087*	-.0093
	(.0042)	(.0044)	(.0061)
day_index=11	-.022***	-.01*	-.0048
	(.0042)	(.0043)	(.0061)
day_index=12	-.023***	-.014**	-.012*
	(.0042)	(.0043)	(.0061)
Day 13 to 28 Omitted for similar results			
day_index=29	-.027***	-.018***	-.017**
	(.0042)	(.0043)	(.006)

day_index=30	-.032***	-.021***	-.02***
	(.0042)	(.0043)	(.006)
Site		-.022***	-.022***
		(.0017)	(.0017)
Price		2.3e-06**	2.3e-06**
		(7.1e-07)	(7.1e-07)
Num Images		.0004***	.0004***
		(.000036)	(.000036)
Num Words		-9.4e-06***	-9.4e-06***
		(1.4e-06)	(1.4e-06)
Desc Rating		-.06***	-.06***
		(.012)	(.012)
Service Rating		.022	.022
		(.018)	(.018)
Logistic Rating		-.038**	-.037**
		(.013)	(.013)
day_index=3 × Endorse			.001 (.0088)
day_index=4 × Endorse			.015+ (.0088)
day_index=5 × Endorse			.0028 (.0088)
Day 6 to 30 Omitted			
Constant	.04***	.4***	.4***
	(.0031)	(.055)	(.055)
Observations	29432	24791	24791
Adjusted R2	0.011	0.028	0.028

Notes. (1) *Standard Error* in parentheses. (2) *Num Followers* are in thousands. (3) Significance codes: $^{+}$ $p <$ 0.1, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$